

Module 24.07: Diagnostic Framework — Detecting Underfitting vs. Overfitting

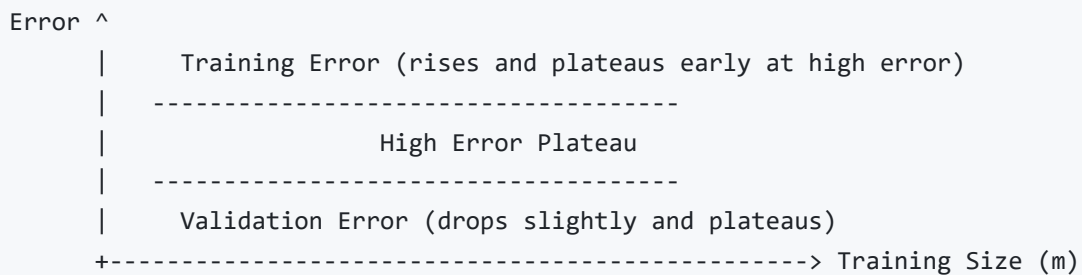
1. The Diagnostic Dilemma

Looking solely at a single test accuracy number is insufficient to determine whether a machine learning model is underfitting or overfitting. A structured diagnostic methodology is required to inspect training dynamics and loss curves.

2. Learning Curves: Training Set Size (m) vs. Error

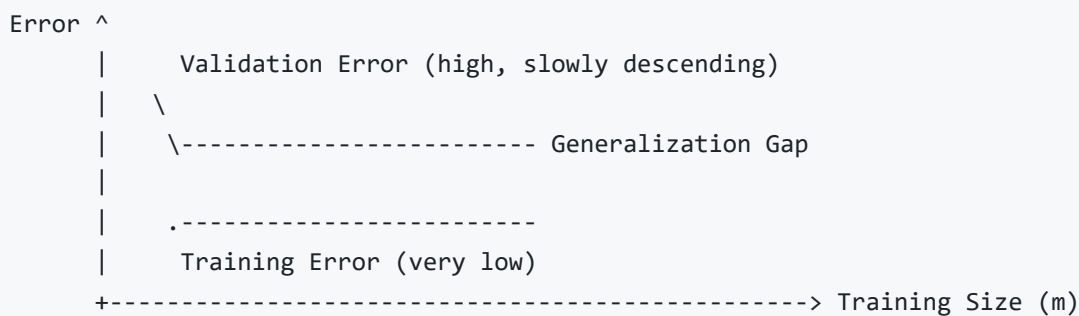
A **learning curve** plots training error and validation error as a function of the training sample size m :

Case A: Underfitting (High Bias) Learning Curve



- **Diagnostic Signature:** Both training and validation errors plateau at a **high** error level.
- **Crucial Rule:** Adding more training data **will NOT help!** The model is already saturated at its maximum capacity.

Case B: Overfitting (High Variance) Learning Curve



- **Diagnostic Signature:** There is a **large gap** between low training error and high validation error.
- **Crucial Rule:** Adding more training data **WILL help!** As m increases, the validation curve continues to converge toward the training curve.

3. Systematic Diagnostic Decision Matrix

Observation	Primary Problem	Prescribed Next Step
$\mathcal{L}_{\text{train}}$ is high, $\mathcal{L}_{\text{val}}$ is high	High Bias (Underfitting)	1. Add more features / interactions 2. Try a more complex model 3. Decrease regularization ($\alpha \downarrow$)
$\mathcal{L}_{\text{train}}$ is low, $\mathcal{L}_{\text{val}}$ is high	High Variance (Overfitting)	1. Collect more training data ($m \uparrow$) 2. Apply / increase regularization ($\alpha \uparrow$) 3. Feature selection (prune noise)
$\mathcal{L}_{\text{train}}$ is low, $\mathcal{L}_{\text{val}}$ is low	Optimal Model	Ready for production deployment